MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HI-FREQUENCY POWER

TYPES 2N5071 and TX2N5071

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a NPN, silicon, VHF, power transistor. The prefix "TX" is used on devices submitted to and passing the special process-conditioning, testing, and screening as specified in 4.5 through 4.5.9.1.

1.2 Physical dimensions. See figure 1 (TO-60).

1.3 Maximum ratings.

P_T 1/ T_A=25°C	P_T 2/ T_C=25°C	V_{CEO}	V_{EBO}	V_{CEX}	I_C	$T_{Oper.}$ & $T_{Stg.}$
W	W	Vdc	Vdc	Vdc	Adc	°C
2.6	70	35	4	65	10	-65 to + 200

1/ Derate lineraly 15 mW/°C for T_A 25°C
2/ Derate lineraly 400 mW/°C for T_C 25°C

1.4 Primary electrical characteristics.

Limits	h_{FE} V_{CE} = 5Vdc I_C = 3 Adc	C_{obo} V_{CB} = 30Vdc I_E = 0 100kHz f 1MHz	P_{out} P_{in} = 3 W f = 76MHz	VSWR f = 30MHz P_O = 30W	θ_{J-C}
		pf	W		°C/W
Min.	15		24		
Max.	100	85	34	3:1 All Phases	2.5

FSC-5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specifications for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices,
MIL-STD-202 - Test Methods for Electronic and Electrical Components Parts.

(Copies of specifications, standards, drawings, and publications required by cuppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS:

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

η - - - - - - - - - - - - - Collector Efficiency ($\eta = \frac{\text{RF Power Out}}{\text{dc Power in}} \times 100$)

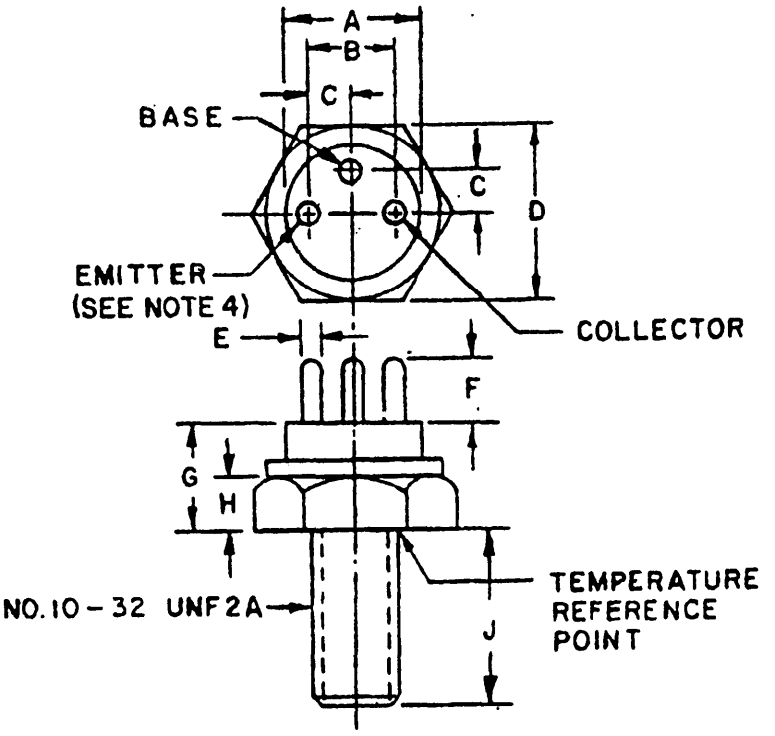
P_{in} - - - - - - - - - - - - - Input Power

P_{out} - - - - - - - - - - - - - Output Power

3.3 Design and construction. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III, and as follows:

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.320	.360	8.13	9.14	
B	.185	.215	4.70	5.46	
C	.090	.110	2.29	2.79	
D	.424	.437	10.77	11.10	
E	.030	.046	.76	1.17	5
F	.140	.160	3.56	4.06	5
G	.215	.320	5.46	8.13	
H	.090	.135	2.29	3.43	
J	.375	.455	9.53	11.56	

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. The pin spacing permits insertion in any socket having a pin circle diameter of .200 inch (50.8 mm) and contacts which will accommodate pins having a diameter of .035 inch (.89 mm) minimum, .045 inch (1.14 mm) maximum.
3. The torque applied to a 10-32 hex nut assembled on the thread during installation should not exceed 12 inch-pounds.
4. Emitter terminal electrically connected to case.
5. All three terminals.

FIGURE 1. Physical dimensions of transistor types 2N5071 and TX2N5071 (TO-60).

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3.4.1 Process-conditioning, testing, and screening for "TX" type. Process-conditioning testing, and screening for the "TX" type shall be as specified in 4.5.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin
(b) Manufacturer's identification

3.5.1 "TX" marking. Devices in accordance with the "TX" requirements shall be marked with "TX" immediately following the "JAN" or "J" prefix.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.2.1 Qualification testing. The non-TX type shall be used for qualification testing. Upon request to the qualifying activity, qualification will be extended to include the TX type of the device.

4.2.2 Table 1 Acceptance Criteria for Qualification Inspection. For qualification inspection, all units that are to be subjected to group B and C inspection shall first be subjected to all tests of group A inspection using a combined LTPD of 5 for group A acceptance criteria.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections. When specified in the contract or order, one copy of the quality conformance inspection data, pertinent to the device inspection lot, shall be supplied with each shipment by the device manufacturer.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.1.1 Load Mismatch test. All transistor shall be subjected to a variable phase VSWR test at 30 MH_z. The P_{out} test fixture shown in Figure 3 shall be adjusted for maximum gain with the transistor under test operating at 30 watts into a 50-ohm load. With the input power then fixed, the 50-ohm load shall then be replaced by a 3:1 mismatch load which automatically traverses all Smith Chart phase variations in a minimum time cycle of 6 seconds. Failure of the transistor to return to 30±1 watts output power when loaded with a 50-ohm termination after one mismatch cycle shall be cause for rejection.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.2.1 Power out end-point test. Collector current shall be recorded and data submitted.

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4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340 hours life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000 hour acceptance evaluation after they have passed the Group B, 340 hour acceptance criteria. The cumulative total of failures found during 340 hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000 hour acceptance criteria.

4.3.5 Group C testing. The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on group C characteristics and parameters for the product concerned. Upon determination by the Government inspector (in advance of group C, 6-month, test results) that group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection, normally for a minimum of 3 consecutive lots, to be performed for required group C tests.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows.

4.4.1 Power output and collector-efficiency measurement. The device shall be tested in the circuit of figure 2 or 3. The specified conditions shall be applied and the variable capacitors adjusted to obtain maximum power output when the maximum power output is obtained, the collector current shall be measured and recorded, the collector efficiency shall be computed as follows:

$$\eta \text{ in } \% = \frac{P_O(\text{watts})}{28 \times I_C(\text{amperes})} \quad X100$$

4.4.2 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.4.3 Resistance to Solvents. Transistors shall be subjected to tests in accordance with Method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied, shall be brushed.

(b) After subjection to the tests there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

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TABLE I. Group A Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	NON TX	TX		Min.	Max.	
Subgroup 1								
Visual and Mechanical Exam.	2071		10	5	---	---	---	---
Subgroup 2			5	3				
Safe Operating area (switching)	3053	Load cond. C; T_A = 25°C; I_C = 200mAdc; L = 25mH, 100 mA; R of inductor = 83 ohms DC resis. (Miller No. 957, or equivalent); R_{BB1} = 150 ohms; V_{BB1} = -20 Vdc; R_{BB2} = Infinity; V_{BB2} = 0; R_S = 1ohm ± 1% 1/2W (non-inductive); V_{CC} = 17Vdc; duty cycle 50% at 60 Hz			BV_{CEO}	35		Vdc
Safe Operating area (switching)	3053	Load cond C; T_A = 25°C; I_C = 200mAdc; L = 25mH, 100mA; R of inductor = 83 ohms DC resis. (Miller No. 957, or equivalent: R_{BB1} = 150 ohms; V_{BB1} = -20 Vdc; R_{BB2} = 20 ohms; V_{BB2} = -1.5 Vdc; R_S = 1ohm ± 1% 1/2watt (non-inductive); V_{CC} = 17 Vdc; duty cycle 50% at 60 Hz			BV_{CEX}	65		Vdc

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TABLE I. Group A Inspection Cont'd

Examination or Test	MIL-STD-750 Method	MIL-STD-750 Details	LTPD NON TX	LTPD TX	Symbol	Limits Min.	Limits Max.	Unit
Breakdown voltage, emitter to base	3026	Bias cond. D; I_E = 10mAdc			BV_{EBO}	4	--	Vdc
Collector to emitter cutoff current	3041	Bias cond. D; V_{CE} = 30Vdc			I_{CEO}	---	5	mAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 60 Vdc			I_{CBO}	---	5	mAdc
Forward-current transfer ratio	3076	V_{CE} = 5Vdc; I_C = 3Adc; pulsed (see 4.4.2)			h_{FE}	15	100	---
Forward-current transfer ratio	3076	V_{CE} = 5Vdc; I_C = 1Adc; pulsed (see 4.4.2)			h_{FE}	20	100	---
Subgroup 3			5	3				
Open circuit output capacitance	3236	V_{CB} = 30Vdc; I_E = 0; 100kHz $\leq$ f $\leq$ 1 MHz			C_{obo}	---	85	pf
Power output	---	V_{CC} = 24Vdc; P_{in} = 3W; f = 76 MHz (see 4.4.1 and Fig. 2)			P_{out}	24	34	watts
Collector efficiency	---	V_{CC} = 24Vdc; P_{in} = 3W; f = 76 MHz (see 4.4.1 and fig. 2)			η	60%	---	---
Power output	---	V_{CC} = 24Vdc; P_{in} = 2W; f = 30 MHz (see 4.4.1 and fig. 3)			P_{out}	33	43	watts
Collector efficiency	---	V_{CC} = 24Vdc; P_{in} = 2W; f = 30 MHz (see 4.4.1 and fig. 3)			η	60%	---	---

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TABLE I. Group A Inspection Cont'd

Examination or Test	MIL-STD-750		LTPD		Symbol	LIMITS		Unit
	Method	Details	Non TX	TX		Min	Max	
Subgroup 3-Cont'd								
Load Mismatch	---	V_{CC} = 24V, P_O = 30W, f = 30MHz (see 4.3.1.1)			ΔP_{out}		1	watts
Subgroup 4			10	10				
High-temperature operation:		T_A = +150°C						
Collector to emitter cutoff current	3041	Bias cond.D; V_{CB} = 60Vdc			I_{CBO}	---	10	mAdc
Low-temperature operation		T_A = -65°C						
Forward-current transfer ratio	3076	V_{CE} = 5Vdc; I_C = 1Adc; pulsed (see 4.4.2)			h_{FE}	7	---	---

TABLE II. Group B inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX		Min	Max	
Subgroup 1			20	20				
Physical dimensions	2066	(see figure 1)			---	---	---	---
Subgroup 2			15	15				
Solderability	2026				---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond.C, 10 cycles; time at temperature extremes=15 minutes min			---	---	---	---
Thermal shock (glass strain)	1056	Test cond. B			---	---	---	---

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TABLE II. Group B inspection - Cont'd

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX		Min	Max	
Subgroup 2 Cont'd								
Terminal strength (tension)	2036	Test cond.A; weight = 5lbs. application time = 15 sec, each terminal			---	---	---	---
Terminal strength (stud torque)	2036	Test cond.D2; torque = 12 in. -lb; time=15sec.			---	---	---	---
Terminal strength (terminal torque)	2036	Test cond.D1; torque = 6 in.-oz; time = 15 sec.			---	---	---	---
Seal (leak-rate)	1071	Test cond.G or H optional for fine leaks; test cond.A,C,D, or F for gross leaks			---	---	$5x10^{-7}$	atm cc/sec
Moisture resistance	1021	Omit initial conditioning			---	---	---	---
End Points:								
Collector to emitter cutoff current	3041	Bias cond. D; V_{CE} = 30Vdc			I_{CEO}	---	5	mAdc
Power output	---	V_{CC} = 24Vdc; Pin =3W; f = 76MHz (see 4.4.1 and fig. 2)			Pout	24	34	watts
Collector efficiency	---	V_{CC} = 24Vdc; Pin =3W; f = 76 MHz (see 4.4.1 and fig. 2)				60%		

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TABLE II. Group B inspection - Cont'd

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX		Min	Max	
Subgroup 3			15	15				
Shock	2016	Nonoperating; 500G; 1.0msec; 5 blows in each orientation; X1, Y1, Y2, and Z1			---	---	---	---
Vibration, Variable frequency	2056				---	---	---	---
Constant acceleration	2006	20,000 G in each orientation: X1, Y1, Y2 and Z1			---	---	---	---
End points: (same as subgroup 2)								
Subgroup 4			20	20				
Salt atmosphere (corrosion)	1041				---	---	---	---
End points: (same as subgroup 2)								
Subgroup 5			7	C=7				
High-temperature life (nonoperating) (TX types only)	1031	Tstg = +200°C			---	---	---	---
High-temperature life (nonoperating) (Non-TX types only)	1032	Tstg = +200°C; time = 340 hours (see 4.3.4)			---	---	---	---
End Points:								
Collector to emitter cutoff current	3041	Bias cond.D; V_{CE} = 30Vdc			I_{CEO}	---	7	mAdc

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TABLE II. Group B inspection - Cont'd

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX		Min	Max	
Subgroup 5 Cont'd								
Power output	---	V_{CC} = 24Vdc; P_{in} =3W; f = 76MHz (see 4.4.1 and fig. 2)			P_{out}	24	34	watts
Collector efficiency	---	V_{CC} = 24Vdc; P_{in} =3W; f = 76MHz (see 4.4.1 and fig. 2)				60%	---	---
Breakdown voltage, emitter to base	3026	Bias cond. D; I_E = 10 mAdc			ΔBV_{EBO}	---	±20	%change from initial recorded value
Load Mismatch	---	V_{CC} = 24V, P_o = 30V, f = 30MHz (see 4.3.1.1)			ΔP_{out}		1.0	watts
Subgroup 6			7	λ=7				
Steady-state operation life (TX types only)	1026	$100°C \leq T_C \leq 150°C$ V_{CB} = 20Vdc; P_T = $20W + \frac{150°C - T_C}{2.5°C}$			---	---	---	---
Steady-state operation life (Non-TX types only)	1027	$100°C \leq T_C \leq 150°C$ V_{CB} = 20Vdc; P_T = $20\ W + \frac{150°C - T_C}{2.5°C}$ time = 340 hrs (see 4.3.4)			---	---	---	---
End Points: (Same as subgroup 5)								

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TABLE III. Group C inspection

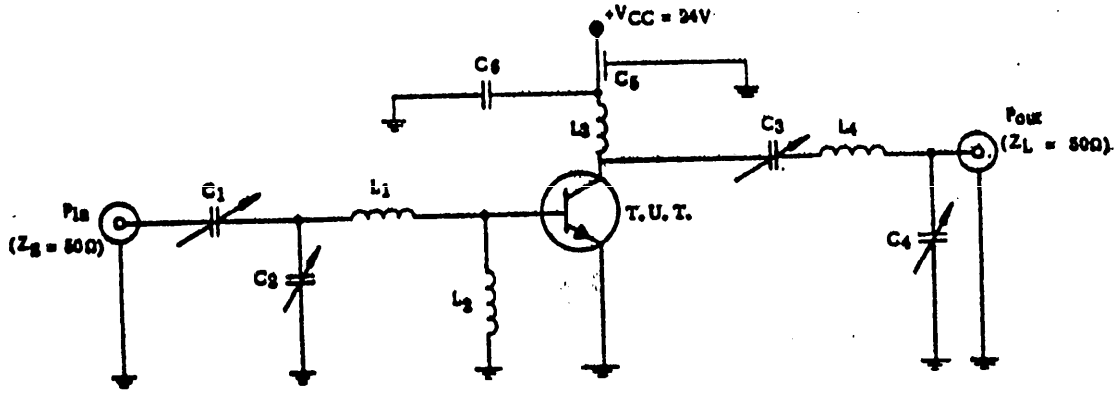
Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX		Min	Max	
Subgroup 1			10	10				
Thermal resistance	3151				θ_{J-C}	---	2.5	°C/W
Safe operating area (continuous DC)	3051	T_A = 25°C t_p = 1 sec for 1 cycle			---	---	---	---
Test #1		V_{CE} = 7Vdc; I_C = 10 Adc						
Test #2		V_{CE} = 15 Vdc I_C = 4.7 Adc						
Test #3		V_{CE} = 35 Vdc I_C = 500 mAdc						
End-points (same as subgroup 2 of Group B)					---	---	---	---
Subgroup 2			20	---				
High-temperature life (nonoperating) (Non-TX types only)	1031	T_{Stg} = +200°C (see 4.3.4)			---	---	---	---
End points: (same as subgroup 5 of group B)								
Subgroup 3			20					
Steady-state operation life (Non-TX types only)	1026	100°C ≤ T_C ≤ 150°C V_{CB} = 20Vdc; P_T = 20 W + $\frac{150°C - T_C}{2.5°C}$ (see 4.3.4)			---	---	---	---
End points: (same as subgroup 5 of group B)								

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TABLE III. Group C inspection Cont'd

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non TX	TX		Min	Max	
Subgroup 4			10	10	---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond.C, 25 cycles; time at temperature extremes = 15 minutes min.						
End points: (same as subgroup 2 of group B)								
Subgroup 5			10	10	---	---	---	---
Resistance to solvents	---	MIL-STD-202, method 215 (see 4.4.3)						

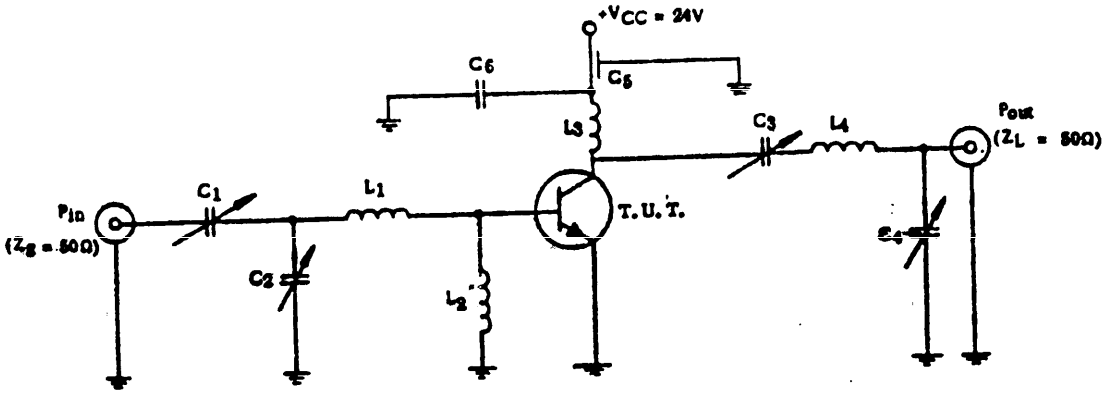
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C_1, C_2: 55-300pf, ARCO 427 or equivalent
C_3, C_4: 32-250pf, ARCO 426 or equivalent
C_5 : 1000pf
C_6 : 0.1μf (50V)
L_1 : 1 turn #16 wire, 5/16 I D
L_2 : Ferrite choke Z = 750Ω
L_3, L_4: 3 turns, #10 wire, 5/16 I D, ¼" long

FIGURE 2. RF amplifier circuit for power output test (76 MHz operation).

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C_1 : 140-680pf, ARCO 468 or equivalent
C_2 : 170-780pf, ARCO 469 or equivalent
C_3 : 24-200pf, ARCO 425 or equivalent
C_4 : 90-400pf, ARCO 429 or equivalent
C_5 : 1000pf
C_6 : 0.1 μf (50V)
L_1 : 2 turns #12 wire ¼ I D, 3/8" long
L_2 : Ferrite choke, Z = 750 Ω
L_3 : 6 turns #14 wire 3/8" I D, 3/4" long
L_4 : 5 turns #10 wire 3/4" I D, 3/4" long

FIGURE 3. RF amplifier circuit for power output test (30 MHz operation).

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4.5 Process-conditioning, testing, and screening for "TX" types. The procedure for process-conditioning, testing, and screening for "TX" types shall be in accordance with 4.5.1 through 4.5.9.1 and figure 4. Process-conditioning shall be conducted on 100 percent of the lot, prior to submission of the lot to the tests specified in tables I, II, and III. (At the option of the manufacturer, the non-TX type may be subjected to process-conditioning and testing.)

4.5.1 Quality assurance (lot verification). Quality assurance shall keep lot records for three years minimum, monitor for compliance to the prescribed procedures, and observe that satisfactory manufacturing conditions and records on lots are maintained for these devices. The records shall be available for review by the customer at all times. The quality assurance monitoring shall include, but not be limited to: process-conditioning, testing, and screening. (The conditioning and screening tests performed as standard production tests need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than specified herein and the relative process-conditioning sequence is maintained.)

4.5.2 High-temperature storage. All devices shall be stored for at least 24 hours at a minimum temperature (T_A) of 200°C.

4.5.3 Thermal shock (temperature cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with MIL-STD-750, method 1051, test condition C, except that 10 cycles shall be continuously performed and the time at the temperature extremes shall be 15 minutes, min.

4.5.5 Acceleration. All devices shall be subjected to acceleration test in accordance with MIL-STD-750, method 2006, with the following exceptions: The test shall be performed one time in the Y_1 orientation only, at a peak level of 10,000G, minimum. The one-minute hold-time requirement shall not apply.

4.5.5 Hermetic seal tests. All devices shall be subjected to hermetic seal tests (fine leak followed by gross leak) with test conditions as specified in 4.5.5.1 and 4.5.5.2. Failed devices from either test shall be removed from the lot.

4.5.5.1 Fine-leak test. All devices shall be fine-leak tested in accordance with MIL-STD-750, method 1071, test condition G or H; except the leak-rate rejection criterion shall be 5×10^{-7} cubic centimeters of helium per second when measured at a differential pressure of one atmosphere.

4.5.5.2 Gross-leak test. All devices shall be tested for gross-leaks in accordance with MIL-STD-750, method 1071, test condition A, C, D or F.

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4.5.6 Reverse bias. All devices shall be subjected to reverse bias with the following test conditions and end point measurement:

(a) The collector to base junction shall be reverse-biased at V_{CB} = 24Vdc, I_E = 0, for 48 hours minimum at T_A = +150°C.
(b) At the end of the high-temperature test time, the ambient temperature shall be lowered until T_C = +30±5°C is attained.
(c) Remove the bias voltage and maintain all devices at room ambient temperature.
(d) Within four hours following bias removal, measure I_{CEO} as specified in table IV.

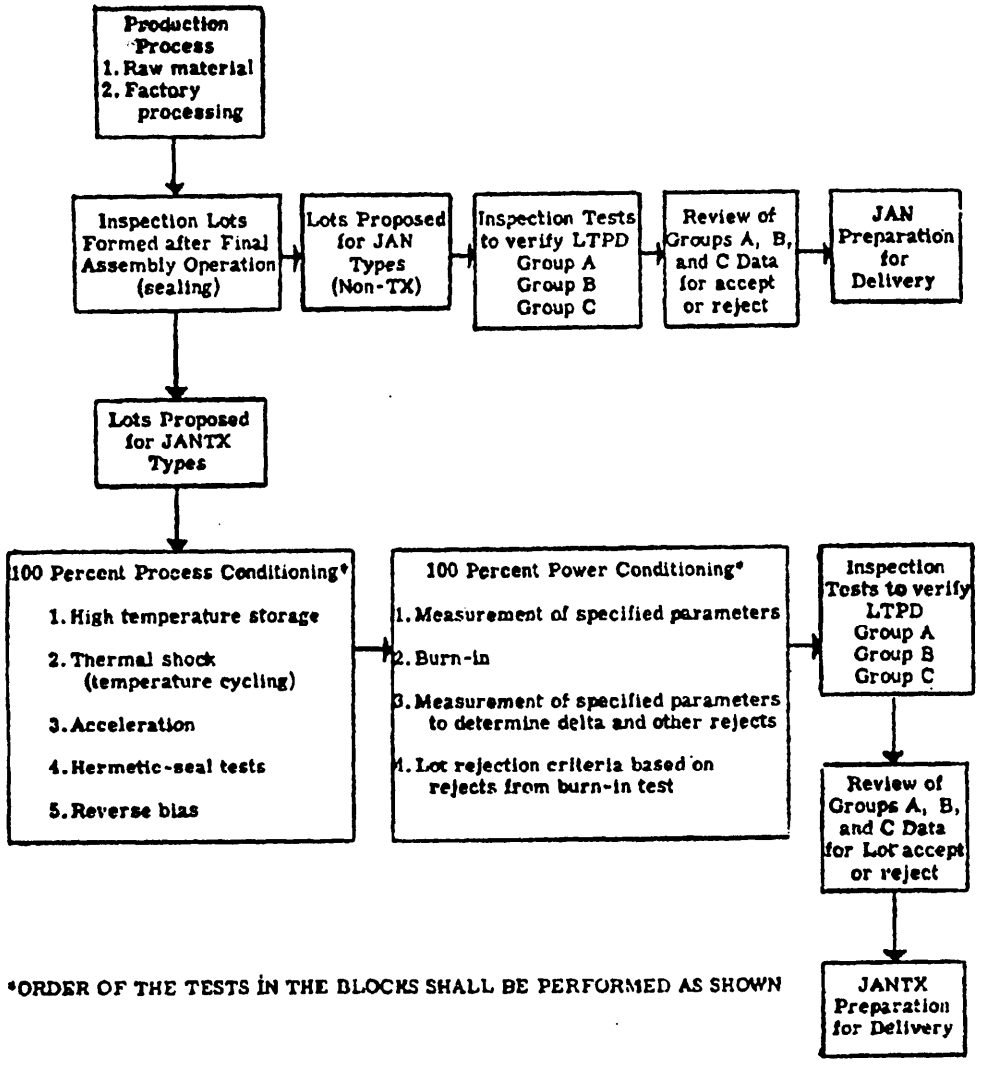


FIGURE 4. Order of procedure diagram for JAN (Non-TX) and JANTX types.

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4.5.7 Preburn-in tests. The parameters I_{CEO} and h_{FE} of table IV shall be measured and the data recorded for all devices in the lot. All devices shall be handled or identified such that the delta end points can be determined after the burn-in test. All devices which fail to meet these requirements shall be removed from the lot and the quantity removed shall be noted on the lot history.

TABLE IV. Burn-in test measurements.

Test	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Collector to emitter cutoff current	3041	Bias cond. D; V_{CE} = 30 Vdc	I_{CEO}	---	5	mAdc
Forward-current transfer ratio	3076	V_{CE} = 5Vdc; I_C = 1Adc; pulsed (see 4.4.2)	h_{FE}	20	100	---

4.5.8 Burn-in test. All devices shall be operated for 168 hours minimum under the following conditions:

$T_A = 25 \pm 3°C$ V_{CB} = 28 Vdc P_T = 2.6 W (No heat sink)

4.5.9 Post burn-in tests. The parameters I_{CEO} and h_{FE} of table IV shall be retested after burn-in and data recorded for all devices in the lot. The parameters measured shall not have changed during the burn-in test from the initial value by more than the specified amount as follows:

ΔI_{CEO} = 100 percent of 0.5 milliamperes, whichever is greater

$\Delta h_{FE} = \pm$ 20 percent

4.5.9.1 Burn-in test failures (screening). All devices that exceed the delta limits of 4.5.9 or the limits of table IV after burn-in, shall be removed from the inspection lot and the quantity removed shall be noted on the lot history. If the quantity removed after burn-in should exceed 10 percent of the number of devices subjected to the burn-in test, the entire inspection lot shall be unacceptable for the "TX" types.

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5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Inspection data (see 4.3).

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